PTH-RPM

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Pacemaker Telemetry Hub for Class III Implantable Devices

Medical Telemetry / Remote Patient Monitoring | Rev A2 - Controlled Review Baseline Program Date TBC | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. Bedside telemetry hub connecting BLE-enabled pacemakers to hospital and remote-monitoring networks while preserving a strictly read-only implant boundary. The architecture combines medical-grade isolated power, a multi-radio BLE/LTE/Wi-Fi SoC, store-and-forward telemetry and a 24-hour battery-backup mission. It shall never acquire therapy-programming authority.

Engineering baseline. The current software-safety allocation places telemetry queueing and recovery within an IEC 62304 Class B lifecycle scope; that allocation requires formal program approval before external release. Cybersecurity uses protected device identity, end-to-end encrypted communications, credential lifecycle controls and the baseline no-inbound-connection posture. Manufacturing provides medical traceability and secure pairing-key provisioning; lifecycle controls cover fail-safe OTA recovery, certificate renewal, battery health and component obsolescence.

Implant BLE profile, radio module, hospital network, backend schema, secure-storage element, battery pack, enclosure, power supply and cleaning process are controlled variants. Exact RF performance, queue depth, clinical latency, battery aging margin and regulatory classification remain program-controlled unless explicitly released. The bedside hub regulatory classification is program-controlled and shall be approved separately from the Class III classification of the implantable pacemaker before external release.

Definition	Controlled product interpretation
Product class	Pacemaker Telemetry Hub for Class III Implantable Devices.
Safety / assurance	Bedside telemetry hub connecting BLE-enabled pacemakers to hospital and remote-monitoring networks while preserving a strictly read-only implant boundary.
Documentation baseline	Program Date TBC - Rev A2 controlled review baseline; external release is blocked until the program approves the documentation baseline date.
Primary architecture	The current software-safety allocation places telemetry queueing and recovery within an IEC 62304 Class B lifecycle scope; that allocation requires formal program approval before external release. Cybersecurity uses protected device identity, end-to-end encrypted communications, credential lifecycle controls and the baseline no-inbound-connection posture. Manufacturing provides medical traceability and secure pairing-key provisioning; lifecycle controls cover fail-safe OTA recovery, certificate renewal, battery health and component obsolescence.
Release philosophy	Implant BLE profile, radio module, hospital network, backend schema, secure-storage element, battery pack, enclosure, power supply and cleaning process are controlled variants. Exact RF performance, queue depth, clinical latency, battery aging margin and regulatory classification remain program-controlled unless explicitly released. The bedside hub regulatory classification is program-controlled and shall be approved separately from the Class III classification of the implantable pacemaker before external release.

PRODUCT GOVERNANCE, READ-ONLY AUTHORITY & REQUIREMENT QUALITY

The hub shall separate telemetry acquisition, storage, networking, update and service functions from any pacemaker therapy-control authority.

If a software, radio or maintenance function attempts to create a therapy command path, the system shall reject the path, record the security/safety event and retain read-only telemetry behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-001	The hub shall separate telemetry acquisition, storage, networking, update and service functions from any pacemaker therapy-control authority.	architecture review, authority matrix and read-only penetration/fault-injection evidence
PTHRPM-002	If a software, radio or maintenance function attempts to create a therapy command path, the system shall reject the path, record the security/safety event and retain read-only telemetry behavior.	architecture review, authority matrix and read-only penetration/fault-injection evidence
PTHRPM-003	A change to therapy interface, radio stack, service access or backend integration shall reopen architecture review, authority matrix and read-only penetration/fault-injection evidence before production release.	architecture review, authority matrix and read-only penetration/fault-injection evidence

Verification, margin and future-change rule

Future variants affecting therapy interface, radio stack, service access or backend integration are screened against the mechanism on this page. The release owner may retain prior evidence only where architecture review, authority matrix and read-only penetration/fault-injection evidence confirms the similarity boundary.

SYSTEM BOUNDARY, PACEMAKER / HOSPITAL / CLOUD OWNERSHIP

The hub shall define its boundary across bedside hardware, pacemaker BLE telemetry, hospital network and remote monitoring backend without assuming ownership of implant therapy.

If an external system provides ambiguous ownership for pairing, telemetry acknowledgement or patient association, the product shall quarantine the ambiguous transaction and prevent it from changing implant state or trusted patient association.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-004	The hub shall define its boundary across bedside hardware, pacemaker BLE telemetry, hospital network and remote monitoring backend without assuming ownership of implant therapy.	system boundary diagram, ICDs and patient/device association review
PTHRPM-005	If an external system provides ambiguous ownership for pairing, telemetry acknowledgement or patient association, the system shall quarantine the ambiguous transaction and prevent it from changing implant state or trusted patient association.	system boundary diagram, ICDs and patient/device association review
PTHRPM-006	Evidence reuse after implant protocol, hospital network, cloud backend or service workflow is permitted only when system boundary diagram, ICDs and patient/device association review demonstrates that the affected assumptions remain valid.	system boundary diagram, ICDs and patient/device association review

Verification, margin and future-change rule

The change trigger for this page is implant protocol, hospital network, cloud backend or service workflow. The impact record identifies moved assumptions and the exact system boundary diagram, ICDs and patient/device association review needed for closure.

OPERATING MODES, PATIENT WORKFLOW & DEGRADED STATES

The hub shall implement controlled modes for boot, pairing, normal telemetry, WAN outage, battery operation, update, service and fail-safe recovery.

Verification focuses on the point where mode state, prerequisite or transition validity is unknown. The required containment is to remain in or return to a non-authoritative telemetry-safe state and preserve queued data where feasible.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-007	The hub shall implement controlled modes for boot, pairing, normal telemetry, WAN outage, battery operation, update, service and fail-safe recovery.	state-machine review, transition tests and recovery fault injection
PTHRPM-008	The design shall detect mode state, prerequisite or transition validity is unknown and shall remain in or return to a non-authoritative telemetry-safe state and preserve queued data where feasible.	state-machine review, transition tests and recovery fault injection
PTHRPM-009	Workflow, update process, radio role or service mode shall be treated as a controlled engineering change; approval requires state-machine review, transition tests and recovery fault injection.	state-machine review, transition tests and recovery fault injection

Verification, margin and future-change rule

Changes involving workflow, update process, radio role or service mode are engineering events, not editorial updates; approval depends on the targeted state-machine review, transition tests and recovery fault injection listed here.

MEDICAL MAINS INPUT, ISOLATION & PROTECTION

The mains power architecture shall provide the isolation and protective functions allocated by the medical electrical safety concept.

If mains fault, protection failure or isolation diagnostic abnormality is detected, the system shall remove or isolate the affected power path while maintaining the safest available telemetry/battery state without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-010	The mains power architecture shall provide the isolation and protective functions allocated by the medical electrical safety concept.	medical electrical safety analysis, dielectric/leakage tests and power-fault DVP
PTHRPM-011	Loss or corruption involving mains fault, protection failure or isolation diagnostic abnormality is detected shall not permit silent normal operation; the system shall remove or isolate the affected power path while maintaining the safest available telemetry/battery state.	medical electrical safety analysis, dielectric/leakage tests and power-fault DVP
PTHRPM-012	Production equivalence following mains module, isolation barrier, connector, protective component or grounding concept requires closure of medical electrical safety analysis, dielectric/leakage tests and power-fault DVP.	medical electrical safety analysis, dielectric/leakage tests and power-fault DVP

Verification, margin and future-change rule

For mains module, isolation barrier, connector, protective component or grounding concept, the configuration owner documents evidence reuse, new risks and the required medical electrical safety analysis, dielectric/leakage tests and power-fault DVP before sign-off.

24-HOUR BATTERY BACKUP & ENERGY RESERVE

The product shall support the baseline 24-hour battery-backup mission using a controlled battery capacity, load profile and reserve policy.

Available battery energy falls below the reserve required for the declared mission is treated as a diagnosable abnormal condition. The system response is to prioritize essential telemetry, issue a persistent low-energy indication and shed nonessential loads according to policy.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-013	The product shall support the baseline 24-hour battery-backup mission using a controlled battery capacity, load profile and reserve policy.	battery-runtime test, load profile review, aging margin and low-energy state validation
PTHRPM-014	When available battery energy falls below the reserve required for the declared mission, the product shall prioritize essential telemetry, issue a persistent low-energy indication and shed nonessential loads according to policy.	battery-runtime test, load profile review, aging margin and low-energy state validation
PTHRPM-015	The released configuration shall not be extended across battery chemistry, capacity, load profile, power-management firmware or reserve threshold without battery-runtime test, load profile review, aging margin and low-energy state validation.	battery-runtime test, load profile review, aging margin and low-energy state validation

Verification, margin and future-change rule

Any modification to battery chemistry, capacity, load profile, power-management firmware or reserve threshold is accepted only after the affected assumptions are rechecked and battery-runtime test, load profile review, aging margin and low-energy state validation is closed.

SOURCE TRANSFER, CHARGING & BROWNOUT RECOVERY

Mains-to-battery and battery-to-mains transfer shall preserve queue integrity, timekeeping and safe software state without creating an implant command path.

The design prevents silent continuation when supply transfer causes reset, brownout or corrupted storage transaction; it must instead recover through the validated startup path, flag incomplete records and prevent duplicate or fabricated telemetry.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-016	Mains-to-battery and battery-to-mains transfer shall preserve queue integrity, timekeeping and safe software state without creating an implant command path.	source-transfer DVP, brownout injection and storage-consistency review
PTHRPM-017	The released implementation shall recover through the validated startup path, flag incomplete records and prevent duplicate or fabricated telemetry whenever supply transfer causes reset, brownout or corrupted storage transaction.	source-transfer DVP, brownout injection and storage-consistency review
PTHRPM-018	Changes to charger, battery pack, PMIC, hold-up network or brownout threshold shall trigger impact review of affected assumptions and closure of source-transfer DVP, brownout injection and storage-consistency review.	source-transfer DVP, brownout injection and storage-consistency review

Verification, margin and future-change rule

The release baseline remains valid across charger, battery pack, PMIC, hold-up network or brownout threshold only within demonstrated similarity. Otherwise, source-transfer DVP, brownout injection and storage-consistency review is repeated or updated.

MULTI-RADIO SoC, MEMORY & COMPUTE PARTITIONING

The multi-radio SoC shall partition BLE acquisition, WAN/LAN communication, storage and update workloads so resource contention cannot invalidate telemetry integrity.

If CPU, memory or I/O contention exceeds the controlled resource budget, the system shall degrade nonessential services, preserve acquisition/queue integrity and report the resource fault.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-019	The multi-radio SoC shall partition BLE acquisition, WAN/LAN communication, storage and update workloads so resource contention cannot invalidate telemetry integrity.	concurrent-load test, memory/resource analysis and watchdog evidence
PTHRPM-020	If CPU, memory or I/O contention exceeds the controlled resource budget, the system shall degrade nonessential services, preserve acquisition/queue integrity and report the resource fault.	concurrent-load test, memory/resource analysis and watchdog evidence
PTHRPM-021	A change to SoC, memory size, OS services, radio concurrency or container/process allocation shall reopen concurrent-load test, memory/resource analysis and watchdog evidence before production release.	concurrent-load test, memory/resource analysis and watchdog evidence

Verification, margin and future-change rule

Future variants affecting SoC, memory size, OS services, radio concurrency or container/process allocation are screened against the mechanism on this page. The release owner may retain prior evidence only where concurrent-load test, memory/resource analysis and watchdog evidence confirms the similarity boundary.

SECURE STORAGE, ROOT OF TRUST & CRYPTOGRAPHIC KEYS

Device identity, pairing material and update trust anchors shall be stored using the released protected-storage architecture.

If trust-anchor integrity, key access control or secure-storage health cannot be established, the product shall block protected operations that depend on the affected credential and enter controlled recovery.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-022	Device identity, pairing material and update trust anchors shall be stored using the released protected-storage architecture.	secure-storage review, key-access tests and corruption/recovery injection
PTHRPM-023	If trust-anchor integrity, key access control or secure-storage health cannot be established, the system shall block protected operations that depend on the affected credential and enter controlled recovery.	secure-storage review, key-access tests and corruption/recovery injection
PTHRPM-024	Evidence reuse after secure element, key store, cryptographic library or provisioning process is permitted only when secure-storage review, key-access tests and corruption/recovery injection demonstrates that the affected assumptions remain valid.	secure-storage review, key-access tests and corruption/recovery injection

Verification, margin and future-change rule

The change trigger for this page is secure element, key store, cryptographic library or provisioning process. The impact record identifies moved assumptions and the exact secure-storage review, key-access tests and corruption/recovery injection needed for closure.

BLE PACEMAKER TELEMETRY ACQUISITION

BLE acquisition shall accept only telemetry sessions that satisfy the released device identity, protocol and data-integrity checks.

Verification focuses on the point where a BLE frame is malformed, unauthenticated, stale or associated with the wrong implant identity. The required containment is to discard or quarantine the frame, maintain session diagnostics and avoid fabricating replacement clinical data.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-025	BLE acquisition shall accept only telemetry sessions that satisfy the released device identity, protocol and data-integrity checks.	BLE protocol conformance, malformed-frame testing and device-association DVP
PTHRPM-026	The design shall detect a BLE frame is malformed, unauthenticated, stale or associated with the wrong implant identity and shall discard or quarantine the frame, maintain session diagnostics and avoid fabricating replacement clinical data.	BLE protocol conformance, malformed-frame testing and device-association DVP
PTHRPM-027	BLE stack, implant profile, radio firmware, antenna or protocol parser shall be treated as a controlled engineering change; approval requires BLE protocol conformance, malformed-frame testing and device-association DVP.	BLE protocol conformance, malformed-frame testing and device-association DVP

Verification, margin and future-change rule

Changes involving BLE stack, implant profile, radio firmware, antenna or protocol parser are engineering events, not editorial updates; approval depends on the targeted BLE protocol conformance, malformed-frame testing and device-association DVP listed here.

PAIRING, BONDING & SESSION ESTABLISHMENT

Pacemaker pairing shall bind the correct patient, hub and implant identities through the approved clinical workflow without exposing therapy control.

If pairing identity is ambiguous, expired, revoked or inconsistent with the patient record, the system shall stop session establishment, preserve audit evidence and require authorized re-association without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-028	Pacemaker pairing shall bind the correct patient, hub and implant identities through the approved clinical workflow without exposing therapy control.	pairing workflow test, negative identity cases and audit-log review
PTHRPM-029	Loss or corruption involving pairing identity is ambiguous, expired, revoked or inconsistent with the patient record shall not permit silent normal operation; the system shall stop session establishment, preserve audit evidence and require authorized re-association.	pairing workflow test, negative identity cases and audit-log review
PTHRPM-030	Production equivalence following pairing policy, identity source, clinical workflow or implant credential format requires closure of pairing workflow test, negative identity cases and audit-log review.	pairing workflow test, negative identity cases and audit-log review

Verification, margin and future-change rule

For pairing policy, identity source, clinical workflow or implant credential format, the configuration owner documents evidence reuse, new risks and the required pairing workflow test, negative identity cases and audit-log review before sign-off.

LTE BACKHAUL & CELLULAR OUTAGE BEHAVIOR

LTE backhaul shall transport queued telemetry without changing acquisition authority or implant behavior.

Cellular registration, tls session or wan service is unavailable is treated as a diagnosable abnormal condition. The system response is to continue store-and-forward operation within local capacity and expose connectivity state without dropping valid queued telemetry silently.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-031	LTE backhaul shall transport queued telemetry without changing acquisition authority or implant behavior.	WAN outage/recovery test, queue-fill analysis and retransmission verification
PTHRPM-032	When cellular registration, TLS session or WAN service is unavailable, the product shall continue store-and-forward operation within local capacity and expose connectivity state without dropping valid queued telemetry silently.	WAN outage/recovery test, queue-fill analysis and retransmission verification
PTHRPM-033	The released configuration shall not be extended across modem, carrier profile, APN, antenna, TLS stack or backend endpoint without WAN outage/recovery test, queue-fill analysis and retransmission verification.	WAN outage/recovery test, queue-fill analysis and retransmission verification

Verification, margin and future-change rule

Any modification to modem, carrier profile, APN, antenna, TLS stack or backend endpoint is accepted only after the affected assumptions are rechecked and WAN outage/recovery test, queue-fill analysis and retransmission verification is closed.

WI-FI HOSPITAL-LAN CONNECTIVITY

Wi-Fi connectivity shall operate within the hospital segmentation and authentication policy while remaining non-authoritative to implant therapy.

The design prevents silent continuation when network authentication, address assignment or segmentation policy is invalid; it must instead deny hospital-LAN service, retain safe local telemetry behavior and log the connectivity reason.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-034	Wi-Fi connectivity shall operate within the hospital segmentation and authentication policy while remaining non-authoritative to implant therapy.	hospital-network integration test, segmentation review and negative authentication cases
PTHRPM-035	The released implementation shall deny hospital-LAN service, retain safe local telemetry behavior and log the connectivity reason whenever network authentication, address assignment or segmentation policy is invalid.	hospital-network integration test, segmentation review and negative authentication cases
PTHRPM-036	Changes to Wi-Fi chipset, enterprise authentication, hospital VLAN policy or network profile shall trigger impact review of affected assumptions and closure of hospital-network integration test, segmentation review and negative authentication cases.	hospital-network integration test, segmentation review and negative authentication cases

Verification, margin and future-change rule

The release baseline remains valid across Wi-Fi chipset, enterprise authentication, hospital VLAN policy or network profile only within demonstrated similarity. Otherwise, hospital-network integration test, segmentation review and negative authentication cases is repeated or updated.

RF ANTENNAS, COEXISTENCE & DESENSE

The BLE/LTE/Wi-Fi antenna system shall maintain released sensitivity and coexistence performance across approved installation and enclosure variants.

If simultaneous radios or nearby emitters cause unacceptable desense or packet loss, the system shall apply the validated concurrency policy, report degraded link quality and preserve queued data integrity.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-037	The BLE/LTE/Wi-Fi antenna system shall maintain released sensitivity and coexistence performance across approved installation and enclosure variants.	RF coexistence test, antenna characterization and simultaneous-radio DVP
PTHRPM-038	If simultaneous radios or nearby emitters cause unacceptable desense or packet loss, the system shall apply the validated concurrency policy, report degraded link quality and preserve queued data integrity.	RF coexistence test, antenna characterization and simultaneous-radio DVP
PTHRPM-039	A change to antenna, enclosure, cable, radio power, coexistence firmware or mounting shall reopen RF coexistence test, antenna characterization and simultaneous-radio DVP before production release.	RF coexistence test, antenna characterization and simultaneous-radio DVP

Verification, margin and future-change rule

Future variants affecting antenna, enclosure, cable, radio power, coexistence firmware or mounting are screened against the mechanism on this page. The release owner may retain prior evidence only where RF coexistence test, antenna characterization and simultaneous-radio DVP confirms the similarity boundary.

PHYSICAL / LOGICAL READ-ONLY ENFORCEMENT

The hub shall contain no released hardware or software path capable of issuing pacemaker therapy-modification commands.

If an interface, parser or maintenance function attempts to access a therapy-control opcode or service, the product shall block the operation at the earliest controlled boundary and record the attempted violation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-040	The hub shall contain no released hardware or software path capable of issuing pacemaker therapy-modification commands.	interface review, negative protocol tests, code analysis and security penetration evidence
PTHRPM-041	If an interface, parser or maintenance function attempts to access a therapy-control opcode or service, the system shall block the operation at the earliest controlled boundary and record the attempted violation.	interface review, negative protocol tests, code analysis and security penetration evidence
PTHRPM-042	Evidence reuse after implant protocol support, diagnostic tool, service firmware or host API is permitted only when interface review, negative protocol tests, code analysis and security penetration evidence demonstrates that the affected assumptions remain valid.	interface review, negative protocol tests, code analysis and security penetration evidence

Verification, margin and future-change rule

The change trigger for this page is implant protocol support, diagnostic tool, service firmware or host API. The impact record identifies moved assumptions and the exact interface review, negative protocol tests, code analysis and security penetration evidence needed for closure.

PROTOCOL PARSER, INPUT VALIDATION & MALFORMED DATA

All implant, hospital and cloud inputs shall be length-, type-, range- and state-validated before they affect trusted telemetry records.

Verification focuses on the point where malformed or contradictory input passes initial transport checks. The required containment is to quarantine the affected record, preserve raw diagnostic context where allowed and prevent it from being presented as validated telemetry.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-043	All implant, hospital and cloud inputs shall be length-, type-, range- and state-validated before they affect trusted telemetry records.	parser fuzzing, boundary tests and data-contract conformance
PTHRPM-044	The design shall detect malformed or contradictory input passes initial transport checks and shall quarantine the affected record, preserve raw diagnostic context where allowed and prevent it from being presented as validated telemetry.	parser fuzzing, boundary tests and data-contract conformance
PTHRPM-045	Message schema, parser library, protocol revision or backend API shall be treated as a controlled engineering change; approval requires parser fuzzing, boundary tests and data-contract conformance.	parser fuzzing, boundary tests and data-contract conformance

Verification, margin and future-change rule

Changes involving message schema, parser library, protocol revision or backend API are engineering events, not editorial updates; approval depends on the targeted parser fuzzing, boundary tests and data-contract conformance listed here.

STORE-AND-FORWARD QUEUE INTEGRITY

The local queue shall preserve ordering, identity, integrity status and delivery state across WAN outage, restart and battery operation.

If queue metadata, record checksum or storage transaction is inconsistent, the system shall isolate the affected record or segment, continue with known-good data and expose a recoverable data-integrity fault without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-046	The local queue shall preserve ordering, identity, integrity status and delivery state across WAN outage, restart and battery operation.	power-loss storage test, queue-wrap test and integrity/recovery analysis
PTHRPM-047	If queue metadata, a record checksum or a storage transaction is inconsistent, the system shall isolate the affected record or segment, continue with known-good data and expose a recoverable data-integrity fault.	power-loss storage test, queue-wrap test and integrity/recovery analysis
PTHRPM-048	Production equivalence following storage device, filesystem, queue algorithm, retention policy or record format requires closure of power-loss storage test, queue-wrap test and integrity/recovery analysis.	power-loss storage test, queue-wrap test and integrity/recovery analysis

Verification, margin and future-change rule

For storage device, filesystem, queue algorithm, retention policy or record format, the configuration owner documents evidence reuse, new risks and the required power-loss storage test, queue-wrap test and integrity/recovery analysis before sign-off.

TIME BASE, TIMESTAMPING & EVENT ORDER

Telemetry timestamps shall derive from controlled time sources and expose uncertainty or unsynchronized state rather than creating false precision.

Trusted time is unavailable, jumps unexpectedly or conflicts across sources is treated as a diagnosable abnormal condition. The system response is to mark affected records with time-quality status and preserve sequence without silently rewriting clinical time.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-049	Telemetry timestamps shall derive from controlled time sources and expose uncertainty or unsynchronized state rather than creating false precision.	time-source failover test, clock-jump injection and ordering analysis
PTHRPM-050	When trusted time is unavailable, jumps unexpectedly or conflicts across sources, the product shall mark affected records with time-quality status and preserve sequence without silently rewriting clinical time.	time-source failover test, clock-jump injection and ordering analysis
PTHRPM-051	The released configuration shall not be extended across RTC, network time source, modem time, firmware or timestamp schema without time-source failover test, clock-jump injection and ordering analysis.	time-source failover test, clock-jump injection and ordering analysis

Verification, margin and future-change rule

Any modification to RTC, network time source, modem time, firmware or timestamp schema is accepted only after the affected assumptions are rechecked and time-source failover test, clock-jump injection and ordering analysis is closed.

TELEMETRY DEDUPLICATION, REPLAY & DELIVERY ACKNOWLEDGEMENT

The delivery pipeline shall distinguish new, duplicate, replayed and previously acknowledged telemetry without deleting the only valid copy.

The design prevents silent continuation when backend acknowledgement is lost or a replayed payload returns after reconnect; it must instead use record identity and delivery state to prevent duplicate clinical interpretation while retaining traceability.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-052	The delivery pipeline shall distinguish new, duplicate, replayed and previously acknowledged telemetry without deleting the only valid copy.	reconnect/replay test, duplicate-injection test and backend contract review
PTHRPM-053	The released implementation shall use record identity and delivery state to prevent duplicate clinical interpretation while retaining traceability whenever backend acknowledgement is lost or a replayed payload returns after reconnect.	reconnect/replay test, duplicate-injection test and backend contract review
PTHRPM-054	Changes to record identifier, acknowledgement protocol, backend retry semantics or queue schema shall trigger impact review of affected assumptions and closure of reconnect/replay test, duplicate-injection test and backend contract review.	reconnect/replay test, duplicate-injection test and backend contract review

Verification, margin and future-change rule

The release baseline remains valid across record identifier, acknowledgement protocol, backend retry semantics or queue schema only within demonstrated similarity. Otherwise, reconnect/replay test, duplicate-injection test and backend contract review is repeated or updated.

WAN OUTAGE, RECONNECT & BACKLOG DRAIN

After connectivity restoration, backlog transmission shall be rate-controlled so acquisition and local safety/status services remain responsive.

If backlog drain saturates CPU, radio or storage bandwidth, the system shall throttle noncritical transfer, preserve live acquisition and expose backlog status until recovery.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-055	After connectivity restoration, backlog transmission shall be rate-controlled so acquisition and local safety/status services remain responsive.	extended-outage test, reconnect burst test and resource-load profiling
PTHRPM-056	If backlog drain saturates CPU, radio or storage bandwidth, the system shall throttle noncritical transfer, preserve live acquisition and expose backlog status until recovery.	extended-outage test, reconnect burst test and resource-load profiling
PTHRPM-057	A change to cellular/Wi-Fi bandwidth, backend ingest policy, queue depth or scheduler shall reopen extended-outage test, reconnect burst test and resource-load profiling before production release.	extended-outage test, reconnect burst test and resource-load profiling

Verification, margin and future-change rule

Future variants affecting cellular/Wi-Fi bandwidth, backend ingest policy, queue depth or scheduler are screened against the mechanism on this page. The release owner may retain prior evidence only where extended-outage test, reconnect burst test and resource-load profiling confirms the similarity boundary.

IEC 62304 CLASS B SOFTWARE LIFECYCLE ALLOCATION

Software items formally allocated by the program to IEC 62304 Class B shall have controlled requirements, architecture, implementation, verification and anomaly records.

If software configuration or anomaly status cannot be reconciled with the released baseline, the product shall block software release and retain the last approved configuration rather than treating the gap as a runtime condition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-058	Software items formally allocated by the program to IEC 62304 Class B shall have controlled requirements, architecture, implementation, verification and anomaly records.	software lifecycle traceability, anomaly review and release checklist
PTHRPM-059	If software configuration or anomaly status cannot be reconciled with the released baseline, software release shall be blocked and the last approved configuration retained rather than treating the gap as a runtime condition.	software lifecycle traceability, anomaly review and release checklist
PTHRPM-060	Evidence reuse after software safety classification, architecture partition, toolchain or lifecycle process is permitted only when software lifecycle traceability, anomaly review and release checklist demonstrates that the affected assumptions remain valid.	software lifecycle traceability, anomaly review and release checklist

Verification, margin and future-change rule

The change trigger for this page is software safety classification, architecture partition, toolchain or lifecycle process. The impact record identifies moved assumptions and the exact software lifecycle traceability, anomaly review and release checklist needed for closure.

TASK SCHEDULING, WATCHDOGS & RESOURCE HEALTH

Runtime supervision shall detect stalled acquisition, queue, radio and update tasks without allowing a failed nonessential task to corrupt trusted telemetry.

Verification focuses on the point where a supervised task misses its timing or health contract. The required containment is to restart or isolate the affected service according to recovery policy and preserve diagnosable state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-061	Runtime supervision shall detect stalled acquisition, queue, radio and update tasks without allowing a failed nonessential task to corrupt trusted telemetry.	watchdog fault injection, stress test and recovery sequencing review
PTHRPM-062	The design shall detect a supervised task misses its timing or health contract and shall restart or isolate the affected service according to recovery policy and preserve diagnosable state.	watchdog fault injection, stress test and recovery sequencing review
PTHRPM-063	Scheduler, watchdog, task period, priority, memory budget or OS component shall be treated as a controlled engineering change; approval requires watchdog fault injection, stress test and recovery sequencing review.	watchdog fault injection, stress test and recovery sequencing review

Verification, margin and future-change rule

Changes involving scheduler, watchdog, task period, priority, memory budget or OS component are engineering events, not editorial updates; approval depends on the targeted watchdog fault injection, stress test and recovery sequencing review listed here.

LOCAL USER INTERFACE, STATUS & PATIENT INDICATIONS

Patient-facing indications shall distinguish power, connectivity, pairing, update and fault states without implying therapy status beyond available telemetry.

If display state is stale, contradictory or cannot be driven, the system shall default to a conservative status indication and preserve service diagnostics without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-064	Patient-facing indications shall distinguish power, connectivity, pairing, update and fault states without implying therapy status beyond available telemetry.	UI state review, stale-data injection and human-factors evidence as applicable
PTHRPM-065	Loss or corruption involving display state is stale, contradictory or cannot be driven shall not permit silent normal operation; the system shall default to a conservative status indication and preserve service diagnostics.	UI state review, stale-data injection and human-factors evidence as applicable
PTHRPM-066	Production equivalence following display, LED, buzzer, status text, localization or UI firmware requires closure of UI state review, stale-data injection and human-factors evidence as applicable.	UI state review, stale-data injection and human-factors evidence as applicable

Verification, margin and future-change rule

For display, LED, buzzer, status text, localization or UI firmware, the configuration owner documents evidence reuse, new risks and the required UI state review, stale-data injection and human-factors evidence as applicable before sign-off.

DIAGNOSTICS, SERVICE LOGS & FIRST-FAULT CAPTURE

Diagnostic logs shall capture power, radio, queue, update and security events with controlled retention and privacy handling.

Logging fails, storage is full or first-fault evidence would be overwritten is treated as a diagnosable abnormal condition. The system response is to preserve the configured critical-event subset, raise service indication and avoid disrupting telemetry acquisition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-067	Diagnostic logs shall capture power, radio, queue, update and security events with controlled retention and privacy handling.	log-capacity test, rollover test and first-fault fault injection
PTHRPM-068	When logging fails, storage is full or first-fault evidence would be overwritten, the product shall preserve the configured critical-event subset, raise service indication and avoid disrupting telemetry acquisition.	log-capacity test, rollover test and first-fault fault injection
PTHRPM-069	The released configuration shall not be extended across log schema, storage allocation, service tool or retention policy without log-capacity test, rollover test and first-fault fault injection.	log-capacity test, rollover test and first-fault fault injection

Verification, margin and future-change rule

Any modification to log schema, storage allocation, service tool or retention policy is accepted only after the affected assumptions are rechecked and log-capacity test, rollover test and first-fault fault injection is closed.

CYBERSECURITY ARCHITECTURE & THREAT BOUNDARIES

The hub shall apply authenticated identities, encrypted communications and least-privilege service boundaries consistent with the program cybersecurity/risk file.

The design prevents silent continuation when a security control reports tamper, integrity failure or unauthorized access; it must instead deny the affected privileged operation, contain the interface and preserve read-only clinical telemetry where safe.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-070	The hub shall apply authenticated identities, encrypted communications and least-privilege service boundaries consistent with the program cybersecurity/risk file.	threat-model review, penetration testing and security-control verification
PTHRPM-071	The released implementation shall deny the affected privileged operation, contain the interface and preserve read-only clinical telemetry where safe whenever a security control reports tamper, integrity failure or unauthorized access.	threat-model review, penetration testing and security-control verification
PTHRPM-072	Changes to network architecture, trust boundary, threat model, cryptographic mechanism or service role shall trigger impact review of affected assumptions and closure of threat-model review, penetration testing and security-control verification.	threat-model review, penetration testing and security-control verification

Verification, margin and future-change rule

The release baseline remains valid across network architecture, trust boundary, threat model, cryptographic mechanism or service role only within demonstrated similarity. Otherwise, threat-model review, penetration testing and security-control verification is repeated or updated.

DEVICE IDENTITY, CERTIFICATES & CREDENTIAL RENEWAL

Hub, backend and approved infrastructure identities shall have controlled issuance, validity, renewal and revocation behavior.

If a certificate is expired, revoked, not yet valid or chains to an untrusted root, the system shall reject the protected connection and preserve local store-and-forward service.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-073	Hub, backend and approved infrastructure identities shall have controlled issuance, validity, renewal and revocation behavior.	certificate lifecycle test, expiry/revocation cases and provisioning audit
PTHRPM-074	If a certificate is expired, revoked, not yet valid or chains to an untrusted root, the system shall reject the protected connection and preserve local store-and-forward service.	certificate lifecycle test, expiry/revocation cases and provisioning audit
PTHRPM-075	A change to certificate authority, validity period, renewal protocol, key algorithm or trust store shall reopen certificate lifecycle test, expiry/revocation cases and provisioning audit before production release.	certificate lifecycle test, expiry/revocation cases and provisioning audit

Verification, margin and future-change rule

Future variants affecting certificate authority, validity period, renewal protocol, key algorithm or trust store are screened against the mechanism on this page. The release owner may retain prior evidence only where certificate lifecycle test, expiry/revocation cases and provisioning audit confirms the similarity boundary.

NO-INBOUND-CONNECTION POLICY & NETWORK SEGMENTATION

The released network design shall enforce the baseline no-inbound-connection policy for externally reachable interfaces, except explicitly approved maintenance channels if any.

If an unsolicited inbound session or prohibited listening service is detected, the product shall refuse the session, log the security event and keep telemetry egress behavior unchanged.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-076	The released network design shall enforce the baseline no-inbound-connection policy for externally reachable interfaces, except explicitly approved maintenance channels if any.	port/service scan, firewall-policy test and segmentation assessment
PTHRPM-077	If an unsolicited inbound session or prohibited listening service is detected, the system shall refuse the session, log the security event and keep telemetry egress behavior unchanged.	port/service scan, firewall-policy test and segmentation assessment
PTHRPM-078	Evidence reuse after OS service, firewall, VPN, hospital network rule or maintenance interface is permitted only when port/service scan, firewall-policy test and segmentation assessment demonstrates that the affected assumptions remain valid.	port/service scan, firewall-policy test and segmentation assessment

Verification, margin and future-change rule

The change trigger for this page is OS service, firewall, VPN, hospital network rule or maintenance interface. The impact record identifies moved assumptions and the exact port/service scan, firewall-policy test and segmentation assessment needed for closure.

SECURE BOOT, FIRMWARE AUTHENTICITY & ROLLBACK CONTROL

Boot firmware shall authenticate executable images and prevent unapproved rollback to known-vulnerable versions.

Verification focuses on the point where image signature, version policy or measured-boot state is invalid. The required containment is to prevent normal application startup and enter the validated recovery path.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-079	Boot firmware shall authenticate executable images and prevent unapproved rollback to known-vulnerable versions.	secure-boot negative tests, rollback test and recovery verification
PTHRPM-080	The design shall detect image signature, version policy or measured-boot state is invalid and shall prevent normal application startup and enter the validated recovery path.	secure-boot negative tests, rollback test and recovery verification
PTHRPM-081	Bootloader, signing key, firmware layout, secure element or version policy shall be treated as a controlled engineering change; approval requires secure-boot negative tests, rollback test and recovery verification.	secure-boot negative tests, rollback test and recovery verification

Verification, margin and future-change rule

Changes involving bootloader, signing key, firmware layout, secure element or version policy are engineering events, not editorial updates; approval depends on the targeted secure-boot negative tests, rollback test and recovery verification listed here.

FAIL-SAFE OTA UPDATE & RECOVERY

OTA update shall preserve a recoverable previous image or equivalent recovery path and shall not create an implant therapy command capability.

If download, installation, power loss or post-update health check fails, the system shall rollback or remain in recovery while preserving the safest available telemetry and audit state without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-082	OTA update shall preserve a recoverable previous image or equivalent recovery path and shall not create an implant therapy command capability.	interrupted-update test, rollback test and post-update regression
PTHRPM-083	If download, installation, post-update health check or recovery after power loss fails, the system shall roll back or remain in recovery while preserving the safest available telemetry and audit state.	interrupted-update test, rollback test and post-update regression
PTHRPM-084	Production equivalence following update agent, image format, partition layout, signing workflow or remote campaign policy requires closure of interrupted-update test, rollback test and post-update regression.	interrupted-update test, rollback test and post-update regression

Verification, margin and future-change rule

For update agent, image format, partition layout, signing workflow or remote campaign policy, the configuration owner documents evidence reuse, new risks and the required interrupted-update test, rollback test and post-update regression before sign-off.

PATIENT DATA AT REST, PRIVACY & DATA MINIMIZATION

Stored telemetry and patient association data shall be limited to the approved purpose and protected according to the program privacy/security requirements.

Data-at-rest protection, access control or patient-association integrity cannot be established is treated as a diagnosable abnormal condition. The system response is to deny unauthorized access, quarantine ambiguous records and preserve audit evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-085	Stored telemetry and patient association data shall be limited to the approved purpose and protected according to the program privacy/security requirements.	data inventory, access-control test, storage encryption verification and privacy review
PTHRPM-086	When data-at-rest protection, access control or patient-association integrity cannot be established, the product shall deny unauthorized access, quarantine ambiguous records and preserve audit evidence.	data inventory, access-control test, storage encryption verification and privacy review
PTHRPM-087	The released configuration shall not be extended across stored fields, retention period, encryption mechanism, service role or patient-association workflow without data inventory, access-control test, storage encryption verification and privacy review.	data inventory, access-control test, storage encryption verification and privacy review

Verification, margin and future-change rule

Any modification to stored fields, retention period, encryption mechanism, service role or patient-association workflow is accepted only after the affected assumptions are rechecked and data inventory, access-control test, storage encryption verification and privacy review is closed.

HOSPITAL / RPM BACKEND DATA CONTRACT

The backend interface shall define telemetry schema, identity, acknowledgement, error handling and version compatibility without transferring implant therapy authority to the hub.

The design prevents silent continuation when backend schema or version is incompatible with the released hub contract; it must instead hold or reject incompatible records with explicit diagnostics rather than silently remapping clinical data.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-088	The backend interface shall define telemetry schema, identity, acknowledgement, error handling and version compatibility without transferring implant therapy authority to the hub.	API/ICD conformance, version-compatibility test and negative schema cases
PTHRPM-089	The released implementation shall hold or reject incompatible records with explicit diagnostics rather than silently remapping clinical data whenever backend schema or version is incompatible with the released hub contract.	API/ICD conformance, version-compatibility test and negative schema cases
PTHRPM-090	Changes to backend API, schema, acknowledgment behavior, cloud endpoint or hospital integration shall trigger impact review of affected assumptions and closure of API/ICD conformance, version-compatibility test and negative schema cases.	API/ICD conformance, version-compatibility test and negative schema cases

Verification, margin and future-change rule

The release baseline remains valid across backend API, schema, acknowledgment behavior, cloud endpoint or hospital integration only within demonstrated similarity. Otherwise, API/ICD conformance, version-compatibility test and negative schema cases is repeated or updated.

MEDICAL RISK CONTROL, READ-ONLY HAZARDS & SAFE DEGRADATION

Risk controls shall address wrong-patient association, missing/delayed telemetry, false status indication and unintended therapy-command capability according to the program risk-management file.

If a safety-related risk control is unavailable or cannot be verified at runtime, the system shall enter the allocated degraded state, prevent misleading clinical output and retain diagnostic evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-091	Risk controls shall address wrong-patient association, missing/delayed telemetry, false status indication and unintended therapy-command capability according to the program risk-management file.	risk-control traceability, fault injection and residual-risk review
PTHRPM-092	If a safety-related risk control is unavailable or cannot be verified at runtime, the system shall enter the allocated degraded state, prevent misleading clinical output and retain diagnostic evidence.	risk-control traceability, fault injection and residual-risk review
PTHRPM-093	A change to risk analysis, clinical workflow, safety control, interface or alarm allocation shall reopen risk-control traceability, fault injection and residual-risk review before production release.	risk-control traceability, fault injection and residual-risk review

Verification, margin and future-change rule

Future variants affecting risk analysis, clinical workflow, safety control, interface or alarm allocation are screened against the mechanism on this page. The release owner may retain prior evidence only where risk-control traceability, fault injection and residual-risk review confirms the similarity boundary.

LOW-ENERGY DEGRADATION & SERVICE PRIORITY

Low-energy policy shall prioritize telemetry acquisition, queue integrity and critical indications over nonessential UI or background services.

If battery reserve or power budget crosses the controlled low-energy threshold, the product shall shed configured nonessential loads and maintain the longest safe telemetry/queue service possible.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-094	Low-energy policy shall prioritize telemetry acquisition, queue integrity and critical indications over nonessential UI or background services.	battery-depletion test, load-shed validation and recovery test
PTHRPM-095	If battery reserve or power budget crosses the controlled low-energy threshold, the system shall shed configured nonessential loads and maintain the longest safe telemetry/queue service possible.	battery-depletion test, load-shed validation and recovery test
PTHRPM-096	Evidence reuse after battery reserve, load priority, display/radio policy or PMIC configuration is permitted only when battery-depletion test, load-shed validation and recovery test demonstrates that the affected assumptions remain valid.	battery-depletion test, load-shed validation and recovery test

Verification, margin and future-change rule

The change trigger for this page is battery reserve, load priority, display/radio policy or PMIC configuration. The impact record identifies moved assumptions and the exact battery-depletion test, load-shed validation and recovery test needed for closure.

PASSIVE THERMAL DESIGN & BEDSIDE TEMPERATURE LIMITS

The fanless or low-acoustic thermal design shall keep components and user-accessible surfaces within approved limits for mains and battery operation.

Verification focuses on the point where temperature sensing is invalid or a controlled thermal threshold is exceeded. The required containment is to reduce nonessential load or stop the affected function while preserving safe power and telemetry state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-097	The fanless or low-acoustic thermal design shall keep components and user-accessible surfaces within approved limits for mains and battery operation.	thermal characterization, hot-soak test and sensor-fault injection
PTHRPM-098	The design shall detect temperature sensing is invalid or a controlled thermal threshold is exceeded and shall reduce nonessential load or stop the affected function while preserving safe power and telemetry state.	thermal characterization, hot-soak test and sensor-fault injection
PTHRPM-099	Enclosure, SoC, modem, power supply, thermal interface or ambient profile shall be treated as a controlled engineering change; approval requires thermal characterization, hot-soak test and sensor-fault injection.	thermal characterization, hot-soak test and sensor-fault injection

Verification, margin and future-change rule

Changes involving enclosure, SoC, modem, power supply, thermal interface or ambient profile are engineering events, not editorial updates; approval depends on the targeted thermal characterization, hot-soak test and sensor-fault injection listed here.

IEC 60601-1 ISOLATION, LEAKAGE & PROTECTIVE MEASURES

The medical power architecture shall demonstrate the insulation, leakage-current and protective-measure requirements allocated by the applicable IEC 60601-1 program profile.

If a dielectric, leakage or protective-earth requirement is not objectively closed, the system shall block release of the affected hardware variant rather than compensating with software without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-100	The medical power architecture shall demonstrate the insulation, leakage-current and protective-measure requirements allocated by the applicable IEC 60601-1 program profile.	safety laboratory evidence, dielectric/leakage report and configuration traceability
PTHRPM-101	Loss or corruption involving a dielectric, leakage or protective-earth requirement is not objectively closed shall not permit silent normal operation; the system shall block release of the affected hardware variant rather than compensating with software.	safety laboratory evidence, dielectric/leakage report and configuration traceability
PTHRPM-102	Production equivalence following power supply, barrier, PCB spacing, connector, grounding or enclosure material requires closure of safety laboratory evidence, dielectric/leakage report and configuration traceability.	safety laboratory evidence, dielectric/leakage report and configuration traceability

Verification, margin and future-change rule

For power supply, barrier, PCB spacing, connector, grounding or enclosure material, the configuration owner documents evidence reuse, new risks and the required safety laboratory evidence, dielectric/leakage report and configuration traceability before sign-off.

EMC, RADIO COEXISTENCE & MEDICAL ENVIRONMENT

The hub shall maintain essential telemetry behavior under the program electromagnetic immunity/emissions profile while coordinating BLE, LTE and Wi-Fi radios.

Emc stress causes reset, data corruption or persistent loss of telemetry integrity is treated as a diagnosable abnormal condition. The system response is to recover through the validated state machine, flag affected data and avoid false clinical status.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-103	The hub shall maintain essential telemetry behavior under the program electromagnetic immunity/emissions profile while coordinating BLE, LTE and Wi-Fi radios.	EMC/ESD DVP, radio coexistence test and post-stress functional verification
PTHRPM-104	When EMC stress causes reset, data corruption or persistent loss of telemetry integrity, the product shall recover through the validated state machine, flag affected data and avoid false clinical status.	EMC/ESD DVP, radio coexistence test and post-stress functional verification
PTHRPM-105	The released configuration shall not be extended across radio, cable, enclosure, PCB layout, filtering or EMC standard applicability without EMC/ESD DVP, radio coexistence test and post-stress functional verification.	EMC/ESD DVP, radio coexistence test and post-stress functional verification

Verification, margin and future-change rule

Any modification to radio, cable, enclosure, PCB layout, filtering or EMC standard applicability is accepted only after the affected assumptions are rechecked and EMC/ESD DVP, radio coexistence test and post-stress functional verification is closed.

WIPE-DOWN ENCLOSURE, CLEANING & MATERIAL COMPATIBILITY

The bedside enclosure shall withstand the approved cleaning agents and wipe-down process without loss of electrical safety, labeling or RF performance.

The design prevents silent continuation when cleaning exposure damages sealing, label legibility or enclosure insulation; it must instead remove the affected configuration from release/service until material compatibility is restored.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-106	The bedside enclosure shall withstand the approved cleaning agents and wipe-down process without loss of electrical safety, labeling or RF performance.	cleaning compatibility test, material review and post-exposure safety/RF check
PTHRPM-107	The released implementation shall remove the affected configuration from release/service until material compatibility is restored whenever cleaning exposure damages sealing, label legibility or enclosure insulation.	cleaning compatibility test, material review and post-exposure safety/RF check
PTHRPM-108	Changes to enclosure resin, coating, label, cleaning agent or seam design shall trigger impact review of affected assumptions and closure of cleaning compatibility test, material review and post-exposure safety/RF check.	cleaning compatibility test, material review and post-exposure safety/RF check

Verification, margin and future-change rule

The release baseline remains valid across enclosure resin, coating, label, cleaning agent or seam design only within demonstrated similarity. Otherwise, cleaning compatibility test, material review and post-exposure safety/RF check is repeated or updated.

CONNECTORS, SERVICE PORTS & ACCESS CONTROL

External connectors and service ports shall prevent misconnections, unauthorized service actions and accidental creation of an inbound therapy-control path.

If service connection is invalid, unsupported or not authorized, the system shall deny privileged actions, preserve audit logging and keep clinical telemetry functions isolated.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-109	External connectors and service ports shall prevent misconnections, unauthorized service actions and accidental creation of an inbound therapy-control path.	connector/interface review, misuse testing and service-authentication verification
PTHRPM-110	If service connection is invalid, unsupported or not authorized, the system shall deny privileged actions, preserve audit logging and keep clinical telemetry functions isolated.	connector/interface review, misuse testing and service-authentication verification
PTHRPM-111	A change to connector, service tool, debug port, cable or access-control policy shall reopen connector/interface review, misuse testing and service-authentication verification before production release.	connector/interface review, misuse testing and service-authentication verification

Verification, margin and future-change rule

Future variants affecting connector, service tool, debug port, cable or access-control policy are screened against the mechanism on this page. The release owner may retain prior evidence only where connector/interface review, misuse testing and service-authentication verification confirms the similarity boundary.

MANUFACTURING TRACEABILITY & DEVICE GENEALOGY

Each unit shall be traceable to hardware, software, radio, battery, secure-provisioning and test identities needed for complaint and field investigation.

If genealogy is incomplete or inconsistent at final assembly, the product shall block unit release until the traceability record is corrected or dispositioned.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-112	Each unit shall be traceable to hardware, software, radio, battery, secure-provisioning and test identities needed for complaint and field investigation.	manufacturing genealogy audit, serialization test and record reconciliation
PTHRPM-113	If genealogy is incomplete or inconsistent at final assembly, unit release shall be blocked until the traceability record is corrected or formally dispositioned.	manufacturing genealogy audit, serialization test and record reconciliation
PTHRPM-114	Evidence reuse after serial-number scheme, MES, BOM, firmware image, provisioning or test database is permitted only when manufacturing genealogy audit, serialization test and record reconciliation demonstrates that the affected assumptions remain valid.	manufacturing genealogy audit, serialization test and record reconciliation

Verification, margin and future-change rule

The change trigger for this page is serial-number scheme, MES, BOM, firmware image, provisioning or test database. The impact record identifies moved assumptions and the exact manufacturing genealogy audit, serialization test and record reconciliation needed for closure.

SECURE PAIRING-KEY PROVISIONING

Pairing keys and device credentials shall be generated or injected through the approved secure manufacturing process with controlled operator and equipment access.

Verification focuses on the point where credential generation, injection or audit evidence fails. The required containment is to quarantine the affected unit and prevent shipment with uncertain identity material.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-115	Pairing keys and device credentials shall be generated or injected through the approved secure manufacturing process with controlled operator and equipment access.	provisioning audit, negative injection test and key-custody review
PTHRPM-116	The design shall detect credential generation, injection or audit evidence fails and shall quarantine the affected unit and prevent shipment with uncertain identity material.	provisioning audit, negative injection test and key-custody review
PTHRPM-117	Provisioning station, HSM, key ceremony, certificate profile or manufacturing site shall be treated as a controlled engineering change; approval requires provisioning audit, negative injection test and key-custody review.	provisioning audit, negative injection test and key-custody review

Verification, margin and future-change rule

Changes involving provisioning station, HSM, key ceremony, certificate profile or manufacturing site are engineering events, not editorial updates; approval depends on the targeted provisioning audit, negative injection test and key-custody review listed here.

EOL RADIO, ANTENNA & CONNECTIVITY TEST

EOL shall verify the released BLE, LTE and Wi-Fi hardware paths at controlled test points without exposing production credentials.

If radio calibration or connectivity evidence is outside the approved acceptance window, the system shall reject or route the unit for controlled rework before secure release without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-118	EOL shall verify the released BLE, LTE and Wi-Fi hardware paths at controlled test points without exposing production credentials.	EOL RF test, calibration limits and traceability review
PTHRPM-119	Loss or corruption involving radio calibration or connectivity evidence is outside the approved acceptance window shall not permit silent normal operation; the system shall reject or route the unit for controlled rework before secure release.	EOL RF test, calibration limits and traceability review
PTHRPM-120	Production equivalence following radio module, antenna, RF path, test fixture or calibration method requires closure of EOL RF test, calibration limits and traceability review.	EOL RF test, calibration limits and traceability review

Verification, margin and future-change rule

For radio module, antenna, RF path, test fixture or calibration method, the configuration owner documents evidence reuse, new risks and the required EOL RF test, calibration limits and traceability review before sign-off.

EOL ISOLATION, BATTERY & 24-HOUR MISSION EVIDENCE

EOL and qualification evidence shall demonstrate power-path safety plus battery-backup capability against the released mission profile and aging assumptions.

Required dielectric, leakage, battery-capacity or transfer evidence is incomplete is treated as a diagnosable abnormal condition. The system response is to block production release of the affected variant until objective evidence is closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-121	EOL and qualification evidence shall demonstrate power-path safety plus battery-backup capability against the released mission profile and aging assumptions.	EOL electrical safety records, battery characterization and transfer test
PTHRPM-122	If required dielectric, leakage, battery-capacity or transfer evidence is incomplete, production release of the affected variant shall be blocked until objective evidence is closed.	EOL electrical safety records, battery characterization and transfer test
PTHRPM-123	The released configuration shall not be extended across battery pack, mains module, charger, isolation barrier or end-of-line process without EOL electrical safety records, battery characterization and transfer test.	EOL electrical safety records, battery characterization and transfer test

Verification, margin and future-change rule

Any modification to battery pack, mains module, charger, isolation barrier or end-of-line process is accepted only after the affected assumptions are rechecked and EOL electrical safety records, battery characterization and transfer test is closed.

END-TO-END TELEMETRY DVP

Product DVP shall exercise implant acquisition, queueing, radio handover, hospital/backend delivery and patient/device association across the released operating envelope.

The design prevents silent continuation when a required DVP condition, dataset or acceptance criterion is missing; it must instead keep the verification gate open and prohibit production-equivalence approval.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-124	Product DVP shall exercise implant acquisition, queueing, radio handover, hospital/backend delivery and patient/device association across the released operating envelope.	DVP matrix, traceability to requirements and objective pass/fail records
PTHRPM-125	The released implementation shall keep the verification gate open and prohibit production-equivalence approval whenever a required DVP condition, dataset or acceptance criterion is missing.	DVP matrix, traceability to requirements and objective pass/fail records
PTHRPM-126	Changes to operating envelope, implant profile, backend version, radio configuration or acceptance criterion shall trigger impact review of affected assumptions and closure of DVP matrix, traceability to requirements and objective pass/fail records.	DVP matrix, traceability to requirements and objective pass/fail records

Verification, margin and future-change rule

The release baseline remains valid across operating envelope, implant profile, backend version, radio configuration or acceptance criterion only within demonstrated similarity. Otherwise, DVP matrix, traceability to requirements and objective pass/fail records is repeated or updated.

OUTAGE, UPDATE & FAULT-INJECTION DVP

Verification shall inject power loss, WAN outage, storage corruption, watchdog expiry, credential failure and interrupted update conditions to demonstrate bounded recovery.

If fault-injection evidence shows silent data loss, false clinical status or unbounded recovery, the system shall fail the affected release gate and retain the last approved configuration.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-127	Verification shall inject power loss, WAN outage, storage corruption, watchdog expiry, credential failure and interrupted update conditions to demonstrate bounded recovery.	fault-injection campaign, recovery timing and data-integrity review
PTHRPM-128	If fault-injection evidence shows silent data loss, false clinical status or unbounded recovery, the system shall fail the affected release gate and retain the last approved configuration.	fault-injection campaign, recovery timing and data-integrity review
PTHRPM-129	A change to fault model, recovery mechanism, update design, storage stack or watchdog shall reopen fault-injection campaign, recovery timing and data-integrity review before production release.	fault-injection campaign, recovery timing and data-integrity review

Verification, margin and future-change rule

Future variants affecting fault model, recovery mechanism, update design, storage stack or watchdog are screened against the mechanism on this page. The release owner may retain prior evidence only where fault-injection campaign, recovery timing and data-integrity review confirms the similarity boundary.

15-YEAR LIFECYCLE, BATTERY / CERTIFICATE SERVICE & OBSOLESCENCE

The lifecycle plan shall track battery health, certificate expiry, storage endurance, radio/SoC obsolescence and supported update paths over the declared 15-year context.

If a component, credential or software dependency reaches an unsupported state, the product shall trigger the approved replacement, renewal or field-update process before loss of required service.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-130	The lifecycle plan shall track battery health, certificate expiry, storage endurance, radio/SoC obsolescence and supported update paths over the declared 15-year context.	lifecycle plan, service thresholds, certificate schedule and obsolescence review
PTHRPM-131	If a component, credential or software dependency reaches an unsupported state, the system shall trigger the approved replacement, renewal or field-update process before loss of required service.	lifecycle plan, service thresholds, certificate schedule and obsolescence review
PTHRPM-132	Evidence reuse after battery supplier, radio module, SoC, certificate policy, OS support or storage device is permitted only when lifecycle plan, service thresholds, certificate schedule and obsolescence review demonstrates that the affected assumptions remain valid.	lifecycle plan, service thresholds, certificate schedule and obsolescence review

Verification, margin and future-change rule

The change trigger for this page is battery supplier, radio module, SoC, certificate policy, OS support or storage device. The impact record identifies moved assumptions and the exact lifecycle plan, service thresholds, certificate schedule and obsolescence review needed for closure.

SUPPLIER CHANGE, VARIANT EQUIVALENCE & FINAL RELEASE CLOSURE

Final release shall require closed safety, cybersecurity, electrical, RF, software, manufacturing and lifecycle evidence for the exact unit configuration.

Verification focuses on the point where supplier or variant evidence is incomplete, contradictory or outside demonstrated similarity. The required containment is to withhold production-equivalence approval until the affected analysis and regression are closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PTHRPM-133	Final release shall require closed safety, cybersecurity, electrical, RF, software, manufacturing and lifecycle evidence for the exact unit configuration.	release checklist, supplier change-impact review and evidence index
PTHRPM-134	If supplier or variant evidence is incomplete, contradictory or outside demonstrated similarity, production-equivalence approval shall be withheld until the affected analysis and regression are closed.	release checklist, supplier change-impact review and evidence index
PTHRPM-135	Supplier, BOM, hardware revision, firmware, manufacturing site or service configuration shall be treated as a controlled engineering change; approval requires release checklist, supplier change-impact review and evidence index.	release checklist, supplier change-impact review and evidence index

Verification, margin and future-change rule

Changes involving supplier, BOM, hardware revision, firmware, manufacturing site or service configuration are engineering events, not editorial updates; approval depends on the targeted release checklist, supplier change-impact review and evidence index listed here.